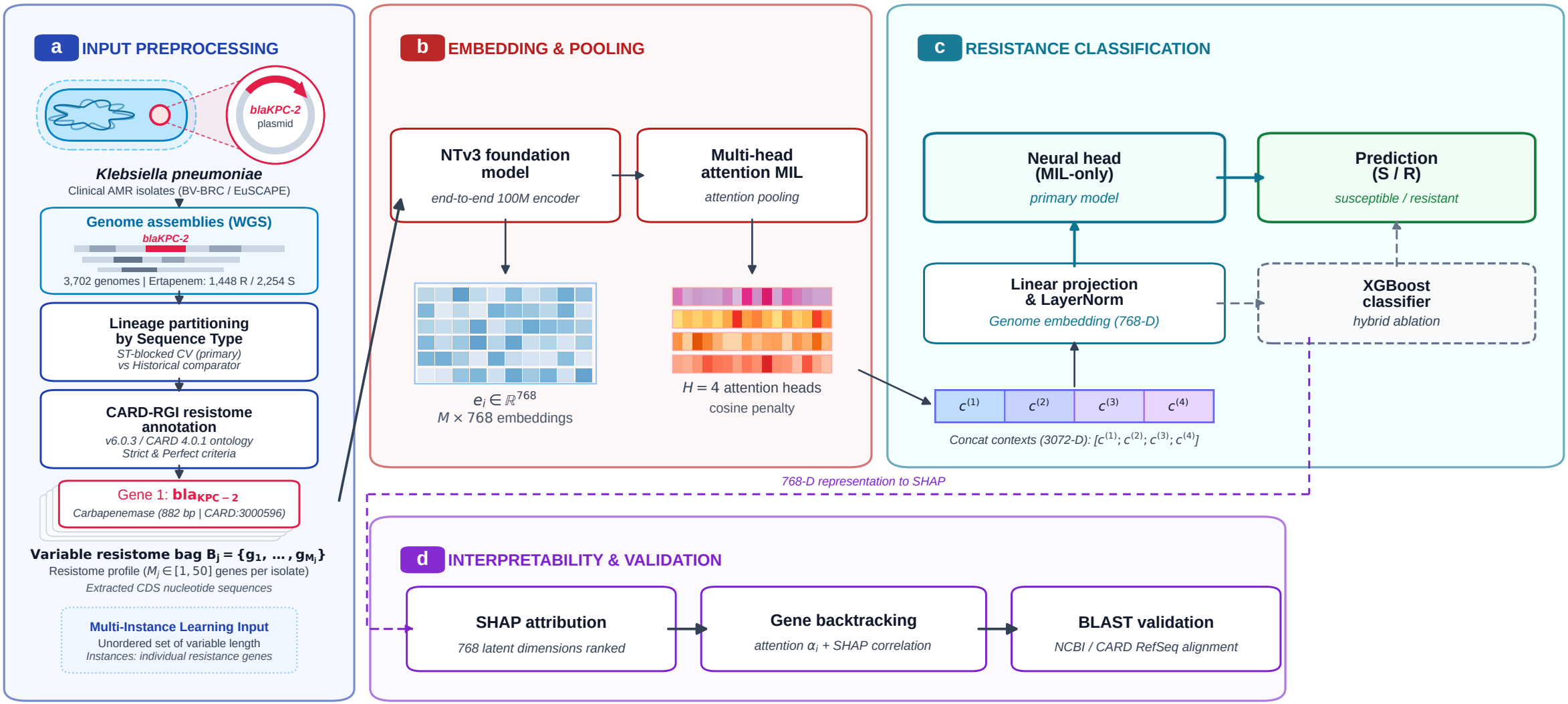

a INPUT PREPROCESSING
blaKPC-2
plasmid
Klebsiella pneumoniae
Clinical AMR isolates (BV-BRC / EuSCAPE)
Genome assemblies (WGS)
blaKPC-2
3,702 genomes | Ertapenem: 1,448 R / 2,254 S
Lineage partitioning by Sequence Type
ST-blocked CV (primary)
vs Historical comparator
CARD-RGI resistome annotation
v6.0.3 / CARD 4.0.1 ontology
Strict & Perfect criteria
Gene 1: blaKPC−2
Carbapenemase (882 bp | CARD:3000596)
Variable resistome bag Bj = {g1, ..., gMj}
Resistome profile (Mj ∈ [1, 50] genes per isolate)
Extracted CDS nucleotide sequences
Multi-Instance Learning Input
Unordered set of variable length
Instances: individual resistance genes
b EMBEDDING & POOLING
NTv3 foundation model
end-to-end 100M encoder
Multi-head attention MIL
attention pooling
ei ∈ ℝ768
M × 768 embeddings
H = 4 attention heads
cosine penalty
c RESISTANCE CLASSIFICATION
Neural head (MIL-only)
primary model
Prediction (S / R)
susceptible / resistant
Linear projection & LayerNorm
Genome embedding (768-D)
XGBoost classifier
hybrid ablation
c(1)
c(2)
c(3)
c(4)
Concat contexts (3072-D): [c(1); c(2); c(3); c(4)]
768-D representation to SHAP
d INTERPRETABILITY & VALIDATION
SHAP attribution
768 latent dimensions ranked
Gene backtracking
attention αi + SHAP correlation
BLAST validation
NCBI / CARD RefSeq alignment